Multi-Agent Collaborative Learning with Multimodal Foundation Models for Cross-Modal AVI Assessment

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Abstract

Asynchronous Video Interviews (AVIs) provide rich multimodal behavioral evidence for assessing candidates' personality traits and cognitive ability. However, effective AVI assessment remains challenging because textual, acoustic, and visual signals exhibit substantial heterogeneity, varying reliability, and different semantic roles across interview questions. Moreover, the limited number of labeled candidates makes end-to-end fine-tuning of large multimodal models prone to overfitting. To address these challenges, we propose a multi-agent collaborative learning framework with multimodal foundation models for cross-modal AVI assessment. Specifically, pretrained foundation models are first employed to extract textual, acoustic, and visual representations, which are subsequently processed by modality-specialized agents. A question-conditioned collaboration module enables the agents to exchange complementary information while preserving modality-specific characteristics. The collaboratively refined representations are then aggregated across multiple interview responses to construct a holistic candidate-level representation. To improve robustness under limited supervision, the foundation-model encoders remain frozen, while only lightweight projection, collaboration, aggregation, and prediction modules are optimized. The proposed framework supports both personality-trait regression and cognitive-ability classification within a unified architecture. Experiments on the AVI Grand Challenge 2026 dataset show that our method achieves an MSE of 0.33892 for personality-trait assessment and an accuracy of 0.58432 for cognitive-ability assessment. These results demonstrate the potential of hierarchical multi-agent collaboration for integrating heterogeneous multimodal evidence in asynchronous video interview analysis.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Empirical studies in HCI**; • **Information systems** → **Information systems applications**.

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1 Introduction

Asynchronous Video Interviews (AVIs) [1] [2] [3] have become an increasingly important component of technology-assisted recruitment, enabling candidates to record responses to predefined questions without requiring interviewers and interviewees to be present simultaneously[4]. Compared with conventional résumé

screening or questionnaire-based assessment, AVIs provide richer behavioral evidence through multiple complementary modalities [5]. The verbal content of a response reflects a candidate's reasoning process, experience, and semantic organization; acoustic patterns convey fluency, confidence, and prosodic variation [6, 7]; and visual behaviors reveal facial expressions [8] [9], gaze, posture, and other nonverbal cues [10]. Effectively integrating these heterogeneous signals has therefore attracted growing interest in personality computing, affective computing [11–13], and automated interview assessment [14] [15] [15].

The AVI Grand Challenge 2026 [16] [17] at ACM Multimedia provides a realistic benchmark for investigating this problem. The challenge contains interview recordings from 644 subjects, each of whom responds to six structured questions, including two generic interview questions and four personality-oriented questions. It consists of two complementary tracks. The first track predicts four self-reported HEXACO personality traits—Honesty-Humility, Extraversion, Agreeableness, and Conscientiousness—from the corresponding interview responses. The second track assesses a candidate's cognitive ability by jointly analyzing all six responses and assigning the candidate to one of three ordinal levels. These two tracks involve different forms of multimodal reasoning: personality assessment emphasizes trait-specific behavioral evidence within individual responses, whereas cognitive ability assessment requires the aggregation of distributed evidence across multiple questions and modalities .

Despite recent progress in multimodal representation learning [18] [19], AVI assessment remains challenging for several reasons [20]. First, the visual, acoustic, and textual modalities have fundamentally different statistical properties and semantic roles. A transcript primarily describes what a candidate says, while audio and visual signals characterize how the response is delivered. Directly concatenating their representations may obscure such modality-specific information and cause high-dimensional features from one modality to dominate the prediction. Second, the reliability of each modality varies considerably across samples. Transcripts may contain automatic speech recognition errors, audio recordings may be affected by noise or silence, and visual streams may contain background clutter, limited facial visibility, or unstable camera conditions. Treating all modalities as equally informative can consequently introduce irrelevant or misleading evidence.[21]

Third, AVI datasets are generally small compared with the scale and dimensionality of modern multimodal foundation models [22]. End-to-end fine-tuning of large video, speech, and language encoders can easily overfit a limited number of candidates. Although frozen pretrained representations provide a practical alternative, their information must still be adapted and coordinated for the target assessment task. Finally, candidate-level assessment cannot be reduced to independent response-level predictions. Different interview questions activate different behaviors and cognitive processes, and a reliable decision should consider both the distinct role of each question and the consistency of evidence across the complete interview. Conventional early-fusion, late-fusion, or single-stream architectures provide only limited mechanisms for modeling these modality-level and question-level interactions.

To address these challenges, we propose *Multi-Agent Collaborative Learning with Multimodal Foundation Models*, a unified framework for cross-modal AVI assessment. As illustrated in Fig. 1, the framework adopts a hierarchical collaborative architecture that first performs question-conditioned multimodal interaction within each interview response and then aggregates evidence across all responses for candidate-level prediction. Instead of forcing heterogeneous information into a single feature stream, we formulate the assessment process as collaboration among three modality-specialized agents. The textual agent analyzes the semantic content and organization of each response using pretrained language representations, the acoustic agent models vocal characteristics and delivery patterns, and the visual agent captures face-centric appearance and nonverbal behavioral cues. Each agent preserves the inductive bias of its corresponding modality while projecting its representation into a shared task space, where a collaboration module learns complementary and consistent cross-modal evidence and reduces the influence of unreliable modality-specific information. To further distinguish the semantic and functional roles of the six interview questions, we incorporate question-aware representations into the modality agents [23]. The resulting response-level representations are subsequently integrated by a candidate-level aggregation module, which models dependencies across questions and constructs a holistic representation of the interviewee. Candidate metadata are finally encoded as supplementary contextual information and fused at the candidate level rather than being directly mixed with raw multimodal features.

Our framework follows a lightweight learning strategy suitable for the limited-data AVI setting. Modality-specific embeddings are extracted using pretrained multimodal foundation models and remain frozen during training, while only the question encoder, modality adaptation modules, collaborative fusion component, metadata encoder, and task-specific prediction heads are optimized. This strategy preserves the transferable knowledge acquired by large-scale pretrained models while substantially reducing the number of trainable parameters and the risk of overfitting. The framework can be instantiated for both response-level personality regression and multi-response cognitive ability classification, providing a unified perspective on the two AVI assessment tasks.

The main contributions of this work are summarized as follows:

- We formulate multimodal AVI assessment as a multi-agent collaborative learning problem, in which modality-specialized agents extract visual, acoustic, and linguistic evidence and jointly reason about the target personality traits or cognitive ability.
- We develop a cross-modal collaboration mechanism that preserves modality-specific information while learning complementary and consistent interactions among heterogeneous foundation-model representations, avoiding the limitations of direct feature concatenation and independent late fusion.
- Extensive experiments on the AVI Grand Challenge 2026 dataset demonstrate the effectiveness of the proposed framework. Our method achieves competitive performance on personality and cognitive ability assessment, validating the benefits of multi-agent cross-modal collaboration for real-world AVI analysis.

2 Related Works

2.1 Automated Assessment from Video Interviews

Automated video interview assessment aims to infer job-related attributes from candidates' recorded responses by analyzing verbal content, vocal delivery, facial behavior, and other observable cues [24]. Early studies mainly relied on manually designed linguistic, acoustic, and visual descriptors, followed by conventional regression or classification models for predicting interview performance, communication skills, personality traits, or hireability. These methods demonstrated that interview videos contain informative behavioral signals beyond résumé-based or questionnaire-based information. However, their performance strongly depended on the quality of handcrafted features and their ability to generalize across recording conditions, questions, and candidate populations [25] [26].

With the development of deep learning, subsequent approaches replaced handcrafted descriptors with representations extracted by convolutional networks, recurrent networks, and Transformer-based architectures. Visual streams have been used to model facial expressions, head movements, gaze, and body behavior, whereas acoustic streams capture prosody, speaking rate, pitch, energy, pauses, and voice quality [27–29]. Textual streams, typically obtained through automatic speech recognition, provide semantic information concerning the relevance, organization, and reasoning expressed in a response [30]. Multimodal systems generally outperform unimodal alternatives because these modalities describe different aspects of interview behavior: textual information reflects what a candidate communicates, while acoustic and visual information characterize how that content is delivered [31].

Existing AVI studies have primarily focused on perceived personality, interview performance, communication competence, or hireability scores. Such targets are often annotated by external observers and may therefore reflect social impressions rather than candidates' underlying psychological attributes. In contrast, the AVI Grand Challenge 2026 [16] emphasizes the assessment of true personality traits using validated self-reported inventories and introduces cognitive ability prediction as a new candidate-level task. This setting requires models to move beyond surface-level impressions and identify behaviorally meaningful evidence distributed across multiple responses. Recent challenge-oriented methods have explored trait-specific predictors, frozen multimodal embeddings, late fusion, and language-model-based semantic analysis. Nevertheless, many of these methods process modalities or interview responses relatively independently, limiting their ability to model structured collaboration across modalities and questions [32].

Another important issue is the reliability and validity of automated AVI systems. Previous studies have shown that interview assessment models may exploit demographic attributes, recording conditions, or other unintended shortcuts instead of task-relevant behavioral evidence. This concern becomes particularly important for cognitive ability assessment, where candidate-level labels are scarce and spurious correlations may dominate the learned decision boundary. Therefore, effective AVI assessment requires not only strong feature extractors but also a structured mechanism that can distinguish complementary task evidence from modality-specific noise and dataset shortcuts [33].

2.2 Multimodal Representation and Fusion Learning

Multimodal learning seeks to integrate information from heterogeneous sources into a unified predictive representation [34, 35]. Conventional fusion strategies can be broadly divided into early fusion, late fusion, and intermediate fusion. Early-fusion methods concatenate modality features before prediction, providing a simple mechanism for joint learning but often ignoring differences in feature distribution, dimensionality, and reliability. Late-fusion methods combine modality-specific predictions and are generally more robust to heterogeneous representations, but they capture only limited fine-grained interactions. Intermediate-fusion approaches instead learn cross-modal interactions inside the network and have become increasingly prominent in multimodal affective and behavioral computing [36, 37].

Representative methods such as the Tensor Fusion Network [38] explicitly model unimodal, bimodal, and trimodal interactions through tensor products. The Multimodal Transformer employs directional cross-modal attention to capture long-range dependencies between unaligned textual, acoustic, and visual sequences. Other approaches use co-attention, graph neural networks, gating mechanisms, or modality-specific expert networks to dynamically exchange and select information. Although these methods improve interaction modeling, they are commonly trained end-to-end on task-specific datasets. Their large numbers of trainable parameters can result in severe overfitting when applied to AVI datasets containing relatively few candidates.

Large-scale pretrained models provide an alternative by transferring knowledge learned from substantially larger external corpora [39]. Models such as CLIP [40] learn generalizable visual representations through image–text contrastive pretraining, while Whisper provides robust speech and transcription representations learned from large-scale weak supervision. Language foundation models further offer semantically rich representations of response content, and multimodal foundation models such as ImageBind demonstrate that heterogeneous modalities can be aligned within a common representation space. These advances make it possible to extract high-quality modality representations without training large encoders directly on the limited AVI data.

However, foundation-model representations [41] are not inherently optimized for psychological or cognitive assessment. Visual encoders pretrained for object recognition may emphasize scene and appearance information rather than subtle interview behavior. Speech encoders may capture phonetic content while overlooking assessment-specific delivery patterns, and language embeddings may prioritize general semantics rather than question-dependent evidence. Directly concatenating these pretrained embeddings therefore does not guarantee effective cross-modal reasoning. Recent AVI methods using frozen foundation models have confirmed their value in small-data settings, but their fusion components are often limited to feature concatenation, score averaging, or independently trained modality predictors. A key remaining

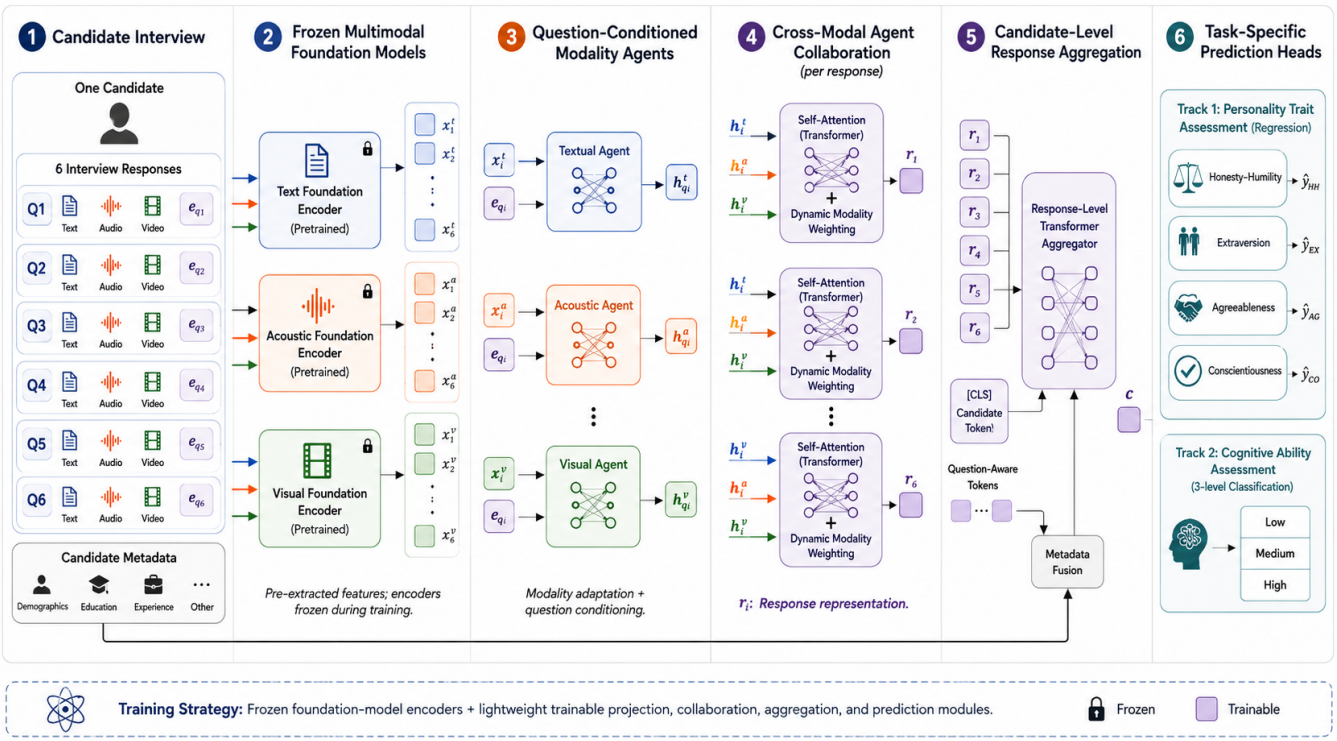


Figure 1: Overview of the proposed multi-agent collaborative learning framework for cross-modal asynchronous video interview assessment.

problem is how to preserve the specialized knowledge of each foundation model while enabling task-oriented information exchange among modalities.

2.3 Multi-Agent Collaborative Learning

Multi-agent learning [42] decomposes a complex problem into multiple interacting agents, each possessing a specialized role, representation, or decision capability. The central motivation is that collaboration among complementary agents can solve problems that are difficult for a single monolithic model. Earlier forms of this idea appear in ensemble learning and mixture-of-experts architectures, where specialized predictors process different subsets or aspects of the input and a gating mechanism combines their outputs. More recently, large-language-model-based frameworks such as CAMEL [43] and AutoGen [44] have investigated role assignment, communication, reflection, and cooperative problem solving among autonomous agents.

Most existing multi-agent studies [45] concentrate on natural-language reasoning, planning, coding, or tool utilization. Their agents commonly communicate through discrete natural-language messages and rely on iterative inference. Such frameworks are not directly suited to AVI assessment, where collaboration must occur over high-dimensional continuous representations and training data are limited. Conversely, conventional multimodal fusion networks rarely formulate individual modalities as autonomous yet jointly optimized agents. They usually treat visual, acoustic, and textual features as passive input streams rather than active participants that independently evaluate evidence and adapt their contributions according to the interview context.

In this work, the term agent refers to a task-oriented modality specialist built upon a corresponding foundation model. Each agent independently encodes and refines evidence from one information source, while retaining its modality-specific inductive bias. Collaboration is then conducted through learnable representation exchange rather than natural-language conversation. This formulation differs from conventional feature fusion in two respects. First, each modality agent generates an assessment-oriented representation before collaboration, preventing its specialized evidence from being overwhelmed by heterogeneous features. Second, the agents are jointly optimized so that their exchanged information is directly guided by the downstream AVI objective.

Moreover, existing multimodal and multi-agent methods generally focus on collaboration among modalities within a single sample. They pay less attention to the hierarchical structure of an interview, in which one candidate provides multiple responses to questions with distinct purposes. Personality-oriented questions may reveal trait-specific behavioral tendencies, whereas generic questions may provide broader evidence of communication and reasoning ability. Simply averaging response representations assumes that all questions carry equivalent information and ignores dependencies across the complete interview. Our framework therefore extends agent collaboration from the modality level to the response and candidate

levels. Modality-specialized agents first collaborate within each response, after which a question-aware coordination mechanism integrates evidence across the interview sequence.

Overall, previous AVI methods have demonstrated the usefulness of multimodal behavioral information, foundation models have provided transferable modality representations, and multi-agent systems have established the benefits of role-specialized collaboration. Nevertheless, these research directions have largely developed independently. Existing approaches do not fully address how modality-specific foundation models can operate as collaborative agents, how their reliability can be adaptively coordinated, and how the resulting evidence can be aggregated across multiple question-conditioned responses. The proposed method bridges these gaps through a hierarchical multi-agent collaborative framework for cross-modal AVI assessment.

3 Proposed Method

3.1 Problem Formulation

Let

$$\mathcal{D} = \{(\mathcal{X}_i, \mathbf{m}_i, \mathbf{y}_i)\}_{i=1}^{N} \tag{1}$$

denote the asynchronous video interview dataset containing N candidates. Each candidate i provides Q interview responses, where $Q = 6$ in the AVI Grand Challenge 2026. The complete interview of candidate i is represented as

$$\mathcal{X}_i = \{X_{i,q}\}_{q=1}^{Q}, \tag{2}$$

where $X_{i,q}$ denotes the response to the q-th interview question.

Each response contains textual, acoustic, and visual modalities:

$$X_{i,q} = \left(X_{i,q}^{t}, X_{i,q}^{a}, X_{i,q}^{v}\right), \tag{3}$$

where t, a, and v denote the textual, acoustic, and visual modalities, respectively. In addition, $\mathbf{m}_i$ denotes the available candidate-level metadata.

The objective is to learn a prediction function

$$f_\Theta : (\mathcal{X}_i, \mathbf{m}_i) \rightarrow \mathbf{y}_i, \tag{4}$$

where $\mathbf{y}_i$ represents either continuous personality-trait scores or a discrete cognitive-ability category. Rather than directly concatenating heterogeneous multimodal features, we formulate AVI assessment as a collaborative learning process among modality-specialized agents.

3.2 Framework Overview

The proposed framework consists of four main components: multimodal foundation-model feature extraction, question-conditioned modality agents, cross-modal collaborative learning, and candidate-level response aggregation.

First, pretrained foundation models extract modality-specific representations from the transcript, audio, and video streams of each response. The foundation models remain frozen during training to reduce overfitting on the relatively small AVI dataset. Next, three lightweight modality agents independently adapt the extracted representations to the target assessment task. A question encoder provides contextual information describing the semantic role of each interview question.

The modality agents subsequently exchange information through a cross-modal collaboration module, producing a unified representation for each response. Finally, the representations of all interview responses are jointly aggregated to construct a candidate-level embedding, which is combined with candidate metadata and passed to a task-specific prediction head.

3.3 Multimodal Foundation-Model Representations

For each response $X_{i,q}$, we employ pretrained foundation models to extract textual, acoustic, and visual representations:

$$\mathbf{x}_{i,q}^{t} = F_t\left(X_{i,q}^{t}\right), \tag{5}$$

$$\mathbf{x}_{i,q}^{a} = F_a\left(X_{i,q}^{a}\right), \tag{6}$$

$$\mathbf{x}_{i,q}^{v} = F_v\left(X_{i,q}^{v}\right), \tag{7}$$

where F_t, F_a, and F_v denote the textual, acoustic, and visual foundation encoders, respectively.

The textual representation captures the semantic content and organization of the candidate's response. The acoustic representation describes speech-related characteristics, such as prosody, fluency, pauses, and vocal delivery. The visual representation captures face-centric appearance and nonverbal behavioral information.

Since the representations extracted by different foundation models may have substantially different dimensionalities and distributions, each modality feature is projected into a shared d-dimensional latent space:

$$\widetilde{\mathbf{x}}_{i,q}^{m} = \phi_m\left(\mathbf{x}_{i,q}^{m}\right), \qquad m \in \{t, a, v\}, \tag{8}$$

where $\phi_m(\cdot)$ is a modality-specific projection network.

The foundation encoders are frozen throughout training, whereas the projection networks and subsequent collaborative modules are optimized for AVI assessment. This strategy preserves transferable knowledge learned from large-scale external datasets while substantially reducing the number of trainable parameters.

3.4 Question-Conditioned Modality Agents

Different interview questions are designed to elicit different behavioral, personality-related, and cognitive evidence. Therefore, treating all responses as semantically equivalent may weaken the question-specific information contained in each answer. To explicitly model the role of each question, we introduce a shared question encoder.

For the q-th interview question, its textual description or question identifier is mapped to a question embedding:

$$\mathbf{e}_q = E_q(q), \tag{9}$$

where $E_q(\cdot)$ denotes the question encoder.

Each modality is modeled as a specialized agent. The input to modality agent m is constructed by combining the projected modality representation with the corresponding question embedding:

$$\mathbf{z}_{i,q}^{m} = A_m\left(\left[\widetilde{\mathbf{x}}_{i,q}^{m}; \mathbf{e}_q\right]\right), \qquad m \in \{t, a, v\}, \tag{10}$$

where $[\cdot;\cdot]$ denotes feature concatenation and $A_m(\cdot)$ is a lightweight modality-specific agent network.

The textual agent focuses on semantic relevance, response organization, and linguistic evidence. The acoustic agent captures speech fluency, prosodic patterns, and vocal delivery. The visual agent models facial and nonverbal behavioral information. By processing each modality independently before collaboration, the framework preserves modality-specific inductive biases and prevents informative signals from being overwhelmed by heterogeneous features.

3.5 Cross-Modal Agent Collaboration

Although each modality agent provides specialized evidence, reliable AVI assessment generally requires information from multiple modalities. We therefore introduce a cross-modal collaboration module that enables the textual, acoustic, and visual agents to exchange and refine their representations.

For response q of candidate i, the three agent representations are organized as a modality-token sequence:

$$\mathbf{Z}_{i,q} = \left[\mathbf{z}^t_{i,q}, \mathbf{z}^a_{i,q}, \mathbf{z}^v_{i,q}\right] \in \mathbb{R}^{3\times d}. \tag{11}$$

A multi-head self-attention layer is used to model interactions among the modality agents:

$$\mathbf{H}_{i,q} = \mathrm{LN}\left(\mathbf{Z}_{i,q} + \mathrm{MHSA}\left(\mathbf{Z}_{i,q}\right)\right), \tag{12}$$

where $\mathrm{MHSA}(\cdot)$ and $\mathrm{LN}(\cdot)$ denote multi-head self-attention and layer normalization, respectively.

The attended representations are further transformed by a feed-forward network:

$$\overline{\mathbf{H}}_{i,q} = \mathrm{LN}\left(\mathbf{H}_{i,q} + \mathrm{FFN}\left(\mathbf{H}_{i,q}\right)\right). \tag{13}$$

Let

$$\overline{\mathbf{H}}_{i,q} = \left[\mathbf{h}^t_{i,q}, \mathbf{h}^a_{i,q}, \mathbf{h}^v_{i,q}\right] \tag{14}$$

denote the collaboratively refined modality representations. Through self-attention, each agent can selectively incorporate complementary evidence from the other modalities. For example, semantic evidence from the transcript can be strengthened by consistent acoustic confidence or facial behavior, while unreliable modality information can receive less attention.

Because the quality and informativeness of each modality may vary across responses, we further estimate a response-dependent reliability score for each modality:

$$s^m_{i,q} = \mathbf{w}_g^\top \tanh\left(\mathbf{W}_g\left[\mathbf{h}^m_{i,q}; \mathbf{e}_q\right] + \mathbf{b}_g\right). \tag{15}$$

The corresponding normalized modality weight is computed as

$$\alpha^m_{i,q} = \frac{\exp\left(s^m_{i,q}\right)}{\sum_{m'\in\{t,a,v\}} \exp\left(s^{m'}_{i,q}\right)}. \tag{16}$$

The response-level multimodal representation is subsequently obtained as

$$\mathbf{r}_{i,q} = \sum_{m\in\{t,a,v\}} \alpha^m_{i,q}\mathbf{h}^m_{i,q}. \tag{17}$$

Unlike fixed feature concatenation or score averaging, the proposed collaboration mechanism jointly learns cross-modal interactions and sample-dependent modality contributions.

3.6 Candidate-Level Response Aggregation

An asynchronous video interview consists of multiple responses rather than a single isolated video. Different questions may reveal complementary aspects of a candidate's personality, communication style, and cognitive ability. We therefore aggregate all response-level representations at the candidate level.

For candidate i, the response representations are arranged as

$$\mathbf{R}_i = \left[\mathbf{r}_{i,1}, \mathbf{r}_{i,2}, \ldots, \mathbf{r}_{i,Q}\right]. \tag{18}$$

We prepend a learnable candidate token $\mathbf{r}_{\mathrm{cls}}$ and inject the corresponding question embeddings:

$$\mathbf{U}^{(0)}_i = \left[\mathbf{r}_{\mathrm{cls}}, \mathbf{r}_{i,1} + \mathbf{e}_1, \ldots, \mathbf{r}_{i,Q} + \mathbf{e}_Q\right]. \tag{19}$$

The resulting token sequence is processed by a response-level Transformer:

$$\mathbf{U}^{(L)}_i = T_{\mathrm{resp}}\left(\mathbf{U}^{(0)}_i\right), \tag{20}$$

where $T_{\mathrm{resp}}(\cdot)$ denotes the response aggregation network and L is its number of layers.

The output corresponding to the candidate token is used as the holistic interview representation:

$$\mathbf{c}_i = \mathbf{U}^{(L)}_{i,0}. \tag{21}$$

This hierarchical architecture models two types of interaction. The cross-modal collaboration module captures relationships among textual, acoustic, and visual evidence within each response, whereas the response aggregation module models dependencies across different interview questions.

3.7 Metadata-Aware Candidate Representation

Candidate-level metadata provide supplementary contextual information that may not be directly observable from individual interview responses. We encode the metadata through a lightweight multilayer perceptron:

$$\mathbf{d}_i = E_m\left(\mathbf{m}_i\right), \tag{22}$$

where $E_m(\cdot)$ denotes the metadata encoder.

The final candidate representation is constructed by combining the multimodal interview representation with the metadata embedding:

$$\mathbf{o}_i = \psi\left(\left[\mathbf{c}_i; \mathbf{d}_i\right]\right), \tag{23}$$

where $\psi(\cdot)$ is a candidate-level fusion network.

Metadata are incorporated only at the candidate level instead of being directly mixed with individual modality features, thereby maintaining a clear separation between behavioral evidence and contextual information.

3.8 Task-Specific Prediction

The proposed framework supports both personality-trait assessment and cognitive-ability classification.

3.8.1 Personality-Trait Assessment. For personality assessment, each target trait is associated with a corresponding personality-oriented interview question. Let $q(p)$ denote the question associated with the p-th personality trait. The predicted score is obtained from its question-conditioned response representation:

$$\widehat{y}^p_i = g_p\left(\mathbf{r}_{i,q(p)}\right), \tag{24}$$

where $g_p(\cdot)$ is a trait-specific regression head.

Using separate prediction heads allows the model to account for the different behavioral manifestations of individual personality dimensions. The personality regression objective is defined as

$$\mathcal{L}_{\text{per}} = \frac{1}{NP} \sum_{i=1}^{N} \sum_{p=1}^{P} \left(\widehat{y}_i^p - y_i^p \right)^2, \tag{25}$$

where P is the number of target personality traits.

3.8.2 Cognitive-Ability Assessment. Cognitive-ability assessment requires the model to jointly consider evidence from all interview responses. The candidate-level representation is therefore passed to a classification head:

$$\widehat{\mathbf{y}}_i^c = \text{softmax}\left(g_c\left(\mathbf{o}_i \right) \right), \tag{26}$$

where $g_c(\cdot)$ is the cognitive-ability classifier.

The corresponding cross-entropy loss is

$$\mathcal{L}_{\text{cog}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{C} y_{i,k}^c \log \widehat{y}_{i,k}^c, \tag{27}$$

where C denotes the number of cognitive-ability categories.

When the personality and cognitive-ability tasks are jointly optimized, the overall objective is defined as

$$\mathcal{L} = \lambda_{\text{per}} \mathcal{L}_{\text{per}} + \lambda_{\text{cog}} \mathcal{L}_{\text{cog}} + \lambda_{\text{reg}} \left\| \Theta \right\|_2^2, \tag{28}$$

where λ_{per}, λ_{cog}, and λ_{reg} control the contributions of personality regression, cognitive classification, and parameter regularization, respectively. When training the model separately for an individual challenge track, only the corresponding task loss is retained.

3.9 Training Strategy

Due to the limited number of labeled candidates, directly fine-tuning all multimodal foundation models may lead to severe overfitting. We therefore adopt a parameter-efficient training strategy. The pretrained textual, acoustic, and visual encoders are used only for feature extraction and remain frozen throughout optimization.

The trainable components include the modality projection layers, shared question encoder, modality-specific agents, cross-modal collaboration module, response aggregation module, metadata encoder, and task-specific prediction heads. This design substantially reduces the number of optimized parameters while allowing the pretrained representations to be adapted to AVI assessment.

Moreover, the separation between foundation-model feature extraction and collaborative learning allows the multimodal embeddings to be extracted and stored before model training, thereby reducing computational and memory costs.

Overall, the proposed framework transforms AVI assessment from conventional multimodal feature fusion into a hierarchical collaborative learning process. Modality-specialized agents first interpret heterogeneous behavioral evidence within each response, after which a candidate-level coordinator integrates information across the complete interview. In this way, the framework preserves modality specialization, exploits cross-modal complementarity, and captures question-dependent evidence for personality and cognitive-ability assessment.

Table 1: Official evaluation results on the AVI Grand Challenge 2026 dataset. A lower MSE is better, while a higher accuracy is better.

Task	Metric	Result
Personality-trait assessment	MSE ↓	0.33892
Cognitive-ability assessment	Accuracy ↑	0.58432

4 Experiments

4.1 Experimental Settings

4.1.1 Dataset and Evaluation Tasks. We evaluate the proposed method on the AVI Grand Challenge 2026 dataset [16], which contains asynchronous video interviews collected from 644 candidates. Each candidate provides six interview responses, including two generic interview questions and four personality-oriented questions. Each response contains textual, acoustic, and visual information, together with candidate-level metadata.

The benchmark consists of two tracks. Track 1 focuses on personality-trait assessment and requires the prediction of four HEXACO personality dimensions, including Honesty–Humility, Extraversion, Agreeableness, and Conscientiousness. Track 2 focuses on cognitive-ability assessment and assigns each candidate to one of three cognitive ability levels based on all six interview responses.

For Track 1, mean squared error (MSE) is adopted as the official evaluation metric, where a lower value indicates better performance. For Track 2, classification accuracy is used, where a higher value indicates better performance.

4.1.2 Implementation Details. All experiments are implemented using PyTorch and conducted on NVIDIA H200 GPUs. The modality-specific representations are extracted before training and remain frozen during optimization. Only the shared question encoder, modality adaptation modules, cross-modal collaboration module, candidate-level aggregation module, metadata encoder, and final prediction heads are trained.

We use AdamW as the optimizer with an initial learning rate of 3×10^{-4} and a weight decay of 10^{-3}. The batch size is set to 32, and the maximum number of training epochs is 200. Early stopping is applied with a patience of 20 epochs according to the validation performance. The gradient clipping threshold is set to 5.0, and a label-smoothing factor of 0.05 is used for the cognitive-ability classification task.

4.2 Overall Performance

The official evaluation results of the proposed method on the AVI Grand Challenge 2026 dataset are summarized in Table 1.

For personality-trait assessment, the proposed method obtains an MSE of 0.33892. This result indicates that the framework can extract trait-related evidence from heterogeneous interview signals and map it to continuous personality scores. In particular, the textual, acoustic, and visual agents focus on different aspects of candidate behavior. The textual agent captures response semantics and organization, the acoustic agent characterizes vocal delivery and speaking patterns, and the visual agent models facial and nonverbal cues. Processing these modalities through specialized agents



Figure 2: Real-world deployment examples of the proposed multimodal safety monitoring system.

before fusion helps preserve modality-specific information that may otherwise be weakened by direct feature concatenation.

The question-conditioned design is also important for personality assessment because different interview questions are associated with different behavioral traits and communicative intentions. By incorporating the corresponding question representation into each modality agent, the model can interpret the same linguistic or behavioral pattern differently according to the context in which it occurs. The subsequent cross-modal collaboration module enables the agents to exchange complementary information and construct a more comprehensive representation of each response. As a result, the prediction is not determined by a single dominant modality, but by the joint evidence conveyed through what the candidate says and how the response is delivered.

For cognitive-ability assessment, the proposed method achieves an accuracy of 0.58432. Compared with personality prediction, this task requires a broader understanding of the complete interview because cognitive evidence may be distributed across all six responses. A candidate may demonstrate reasoning ability, response coherence, verbal organization, or problem-solving behavior in different questions. Therefore, independently predicting from each response or simply averaging response-level decisions may fail to capture the relationships among questions.

The candidate-level aggregation module addresses this issue by jointly modeling the response representations of the entire interview. Through question-aware aggregation, the model preserves the distinct role of each interview question while combining complementary evidence across responses. This mechanism allows the final representation to reflect not only the quality of individual answers, but also the consistency and overall organization of the candidate's performance throughout the interview. The cognitive-ability result therefore supports the use of hierarchical modeling, where multimodal interactions are first learned within each response and cross-question dependencies are subsequently captured at the candidate level.

Another factor contributing to the obtained performance is the use of frozen multimodal foundation models. The challenge dataset contains a relatively limited number of candidates, whereas end-to-end optimization of large textual, acoustic, and visual encoders would introduce a large number of trainable parameters and increase the risk of overfitting. Our framework instead transfers representations learned from large-scale external data and trains only lightweight modality adaptation, collaboration, aggregation, and prediction modules. This parameter-efficient strategy provides a practical balance between representation capability and task-specific adaptation.

The results of the two tracks further illustrate the flexibility of the proposed framework. For Track 1, the model emphasizes question-conditioned response-level representations for continuous personality prediction. For Track 2, it relies on candidate-level aggregation for discrete cognitive-ability classification. Although the two tasks differ in their prediction targets and evidence granularity, they can be addressed within the same hierarchical architecture.

5 Applications

Beyond the experimental evaluation, the proposed Multi-Agent Collaborative Learning with Multimodal Foundation Models for Cross-Modal AVI Assessment has been deployed in a real-world multimodal safety monitoring system developed by the authors' company. In practical scenarios, the proposed framework serves as a core module for automatic personality assessment and collaborates with other multimodal perception algorithms, including MLLM-Based Perception of Risk Behaviors in Indoor Public Spaces etc., to provide comprehensive situation awareness and intelligent risk analysis. An example of the real-world deployment scenario is illustrated in Fig. 2. By integrating cross-modal video understanding, behavior assessment, and risk perception into a unified multimodal pipeline, the system enables timely detection, assessment, and early warning of potential safety hazards in complex environments. The deployment has demonstrated the practicality, scalability, and robustness of the proposed framework in supporting intelligent social governance and public safety protection, illustrating its potential for broader applications in large-scale urban safety management and smart surveillance systems.

6 Conclusion

In this work, we proposed a multi-agent collaborative learning framework with multimodal foundation models for asynchronous video interview assessment. The framework models textual, acoustic, and visual information using modality-specialized agents, performs cross-modal collaboration within each response, and aggregates evidence across multiple interview questions for candidate-level prediction. To reduce overfitting on the limited AVI dataset, the pretrained foundation-model encoders are kept frozen, while only lightweight task-specific modules are optimized. Experiments on the AVI Grand Challenge 2026 dataset show that our method achieves an MSE of 0.33892 for personality-trait assessment and an accuracy of 0.58432 for cognitive-ability assessment. Future work will focus on improving generalization, robustness to missing or unreliable modalities, and fairness in real-world AVI applications.

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